

Harnessing Terminal Signal-Aware Deep Learning for Accurate Multi-Class Secreted Effector Identification

Lesong Wei, Shida He, Quan Zou, Chen Lin

Abstract—Gram-negative bacterial secreted effectors are translocated through specialized secretion systems to manipulate host cellular processes, and their accurate identification is crucial for understanding bacterial pathogenesis. Recent deep learning methods have significantly advanced this field, yet current approaches primarily rely on global sequence representations, overlooking the biological significance of terminal regions where secretion signals reside. Moreover, severe class imbalance among different secreted effector types remains a critical challenge for multi-class prediction. Here, we propose TermSE, a terminal signal-aware framework for multi-class secreted effector identification. TermSE explicitly captures N-terminal and C-terminal sequence features through convolutional neural networks applied to protein language model embeddings, and integrates them with global sequence representations for multi-view sequence characterization. To address class imbalance, TermSE employs a cosine-normalized classifier combined with weighted sampling to mitigate feature magnitude bias and ensure sufficient learning from minority classes. Extensive experiments demonstrate that TermSE outperforms existing methods in both cross-validation and independent test settings, with robust generalization across varying sequence identity levels. Furthermore, interpretability analysis confirms that TermSE learns to focus on biologically meaningful terminal patterns specific to each secreted effector type. These results highlight the potential of TermSE as an effective and interpretable tool for secreted effector discovery.

Index Terms—Secreted effector identification, terminal signal-aware framework, weighted sampling, cosine-normalized classifier.

I. INTRODUCTION

Gram-negative bacteria are major clinical pathogens whose virulence depends on sophisticated protein secretion machinery [1]. These bacteria employ multiple secretion systems, including type I (T1SS), II (T2SS), III (T3SS), IV (T4SS), and VI (T6SS), to translocate effector proteins across their inner and outer membranes, delivering them directly into host cells or the extracellular milieu [2, 3].

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Once translocated, these effectors perform diverse and highly specific functions that systematically interfere with or hijack normal host physiological processes, establishing favorable microenvironments for bacterial replication and survival [4, 5]. For instance, *Yersinia* exploits its T3SS to inject effectors such as YopE and YopH that disrupt host actin filaments, thereby suppressing phagocytosis by macrophages and enabling extracellular persistence [6]. *Legionella pneumophila* utilizes its Dot/Icm T4SS to deliver up to 300 effectors into host macrophages, where they collectively block phagosome-lysosome fusion and establish a replicative compartment, the pathogenic basis of Legionnaires' disease [7, 8]. *Vibrio cholerae* deploys its T6SS to inject phospholipase toxins into competing bacteria or eukaryotic cells, compromising membrane integrity and conferring competitive advantages in complex microbial ecosystems [9]. Given the mechanistic diversity and substrate specificity across secretion systems, accurate identification and classification of secreted proteins are therefore essential for elucidating bacterial pathogenesis and developing next-generation antimicrobial therapeutics.

With growing understanding of bacterial secretion systems, an increasing number of secreted effectors have been identified and characterized across diverse Gram-negative pathogens [10–13]. Several public databases have been developed to curate and integrate experimentally validated effector proteins from the literature. For example, BastionHub [14] offers one of the most comprehensive repositories, integrating known substrates across five major secretion system types in Gram-negative bacteria and notably unifying type I and type II secreted effectors (T1SEs and T2SEs), two categories previously underrepresented in public resources. In addition to such comprehensive resources, system-specific databases such as SecReT4 [15], SecReT6 [16], and BEAN 2.0 [17] compile curated collections of type IV, type VI, and type III secreted effectors (T4SEs, T6SEs, and T3SEs), respectively. Together, these curated datasets provide critical benchmarks for developing computational tools to discover novel secreted effectors.

Experimental discovery of effectors is labor- and time-

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The source code and data used in this study is available at <https://github.com/LesongWei/TermSE>.

consuming, prompting the development of numerous computational methods. Early approaches relied on manually

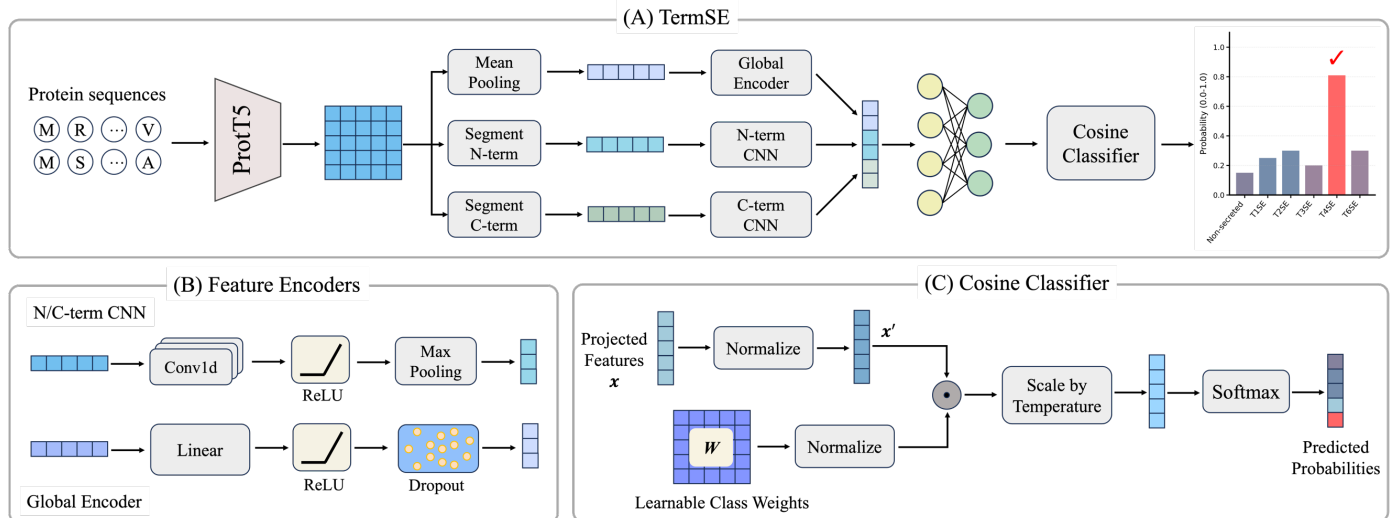


Fig. 1. (A) The overall framework of TermSE, which integrates N-terminal and C-terminal sequence features with global sequence embeddings for six-class secreted effector classification. (B) Architecture of the feature encoders, including the N/C-terminal CNN and global encoder branches. (C) Architecture of the cosine-normalized classifier, which computes similarity between normalized feature vectors and learnable class weight vectors scaled by a temperature parameter.

engineered features, including position-specific scoring matrices (PSSMs), physicochemical properties, and secondary structures, with classifiers designed for specific effector types, including T1SEs [18, 19], T3SEs [20–24], T4SEs [25–29], and T6SEs [30–32]. For instance, Bastion3 [21] employed PSSM-derived features in a two-layer ensemble predictor for T3SEs, while CNN-T4SE [26] integrated convolutional networks to extract hierarchical features from PSSMs for T4SE identification. Bastion6 [30] combined multiple sequence features for T6SE prediction with high accuracy. Despite reasonable performance, these methods were limited to single systems and incurred high computational costs from homology-based alignments, hindering genome-wide scalability. Manually crafted features also struggled to capture effector sequence diversity, potentially missing critical patterns. Thus, while improving identification accuracy, early models remained constrained by poor generalization, low computational efficiency, and limited feature expressiveness.

To address these limitations, recent advances in protein language models (pLMs) have transformed secreted protein identification. Large-scale pLMs, built on Transformer architectures and trained unsupervised on hundreds of millions of sequences, automatically learn complex patterns and capture evolutionary and structural information, eliminating manual feature engineering. pLM embeddings have significantly improved effector classification. For instance, T4SEfinder [29] combined pLM embeddings with PSSM features to achieve state-of-the-art Type IV effector prediction. Recent studies have further adopted multi-class classification strategies, enabling a single model to simultaneously distinguish multiple secretion system types and thereby leverage correlations across categories. DeepSecE [33] achieved multi-class prediction of T1SEs–T4SEs and T6SEs using pLM embeddings and demonstrated attention-based signal peptide localization. MoCETSE [34] further incorporated convolutional expert

networks and relative positional encoding with pLM representations to enhance multi-scale feature extraction and long-range modeling, achieving substantial gains in multi-class prediction. These advances demonstrate that pLM-based approaches overcome manual feature engineering limitations and deliver superior performance and generalization.

However, despite significant performance improvements, existing models still face two critical challenges. First, current approaches typically rely on global sequence representations that treat the entire protein uniformly, overlooking the biological significance of terminal regions. In Gram-negative bacterial secretion systems, substrate recognition is predominantly mediated by specific signal sequences located at the N-terminus or C-terminus [1, 2]. For instance, T2SE [35] and T3SE [36] effectors are known to harbor N-terminal secretion signals, while T4SE [37] substrates are typically recognized through C-terminal translocation motifs. Second, the severe class imbalance among different secretion system types poses a significant obstacle for multi-class prediction. In curated datasets, certain effector types (e.g., T3SEs and T4SEs) are well-represented with hundreds of experimentally validated samples, while others (e.g., T1SEs and T2SEs) remain substantially underrepresented [14], leading to biased models that disproportionately favor majority classes over minority ones.

In this work, we propose TermSE, a terminal signal-aware deep learning framework for accurate multi-class secreted effector identification. Leveraging the representation capacity of pLMs, which capture evolutionary and biochemical patterns from large-scale protein sequences, TermSE explicitly models the biological prior that secretion signals are concentrated at protein termini. Specifically, it extracts localized features from N-terminal and C-terminal regions using 1D convolutional neural networks (CNNs) applied to ProtT5-XL-UniRef50 (ProtT5) [38] embeddings, while simultaneously incorporating

global sequence context from the same pLM representations.

These complementary features are integrated through a unified

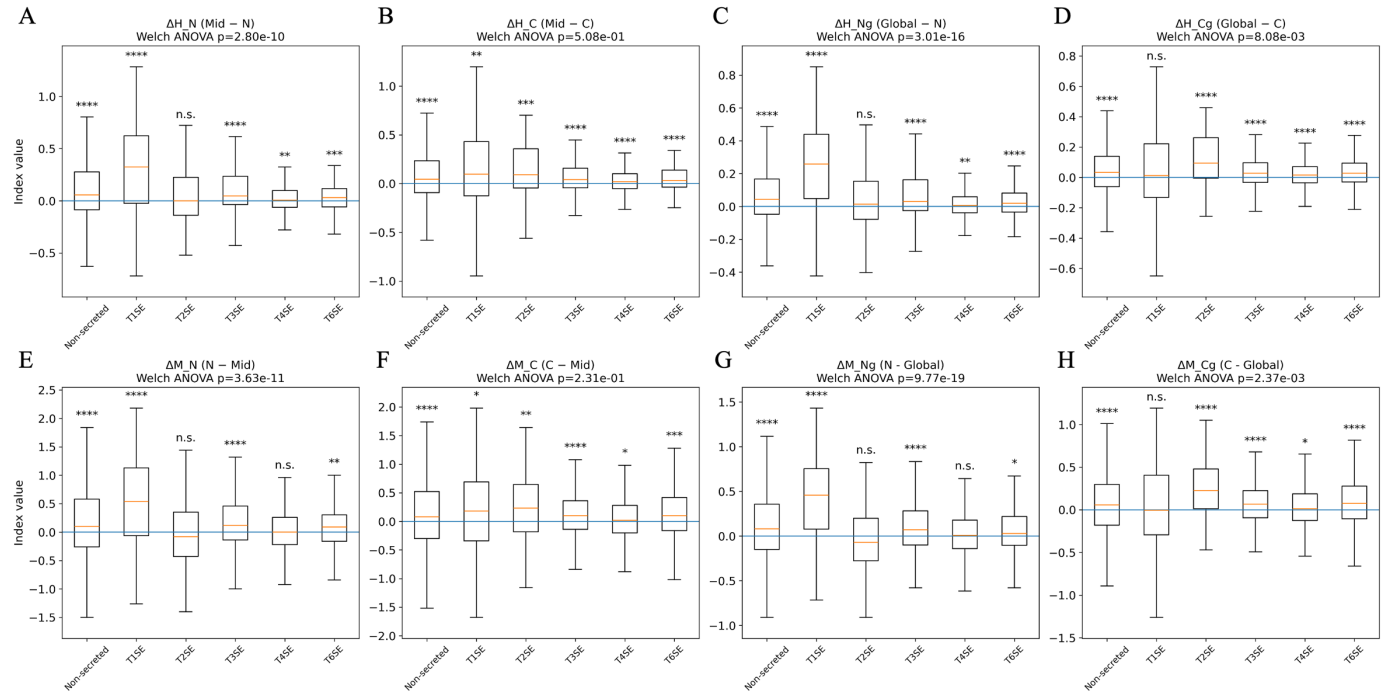


Fig. 2. PSSM-based quantification of terminal signal enrichment across secreted effector types. (A–D) Entropy differences (ΔH) between terminal regions and either the sequence mid-segment (Mid) or the full-length background (Global): ΔH_N (Mid–N) = $H_Mid - H_N$, ΔH_C (Mid–C) = $H_Mid - H_C$, ΔH_Ng (Global–N) = $H_Global - H_N$, and ΔH_Cg (Global–C) = $H_Global - H_C$. (E–H) Maximum score differences (ΔM) between terminal regions and the same two reference regions: ΔM_N (N–Mid) = $M_N - M_Mid$, ΔM_C (C–Mid) = $M_C - M_Mid$, ΔM_Ng (N–Global) = $M_N - M_Global$, and ΔM_Cg (C–Global) = $M_C - M_Global$. Boxplots show the distribution of each index across non-secreted effectors and five secreted effector types. The horizontal line at zero indicates no difference; positive ΔH indicates lower entropy, i.e., stronger positional constraints at termini, while positive ΔM indicates stronger amino acid preferences. Asterisks denote within-class significance against zero by Wilcoxon signed-rank test: n.s., $p \geq 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 10^{-4}$. Panel titles report Welch ANOVA p-values for between-class differences in each index across secreted effector types.

fusion module and fed into a cosine-normalized classifier. Combined with a weighted sampling strategy, this architecture enables robust six-class classification covering five major secreted effector types and non-secreted proteins, while effectively mitigating severe class imbalance. To validate our terminal-focused design, we performed PSSM-based analysis, which demonstrates that secreted proteins exhibit significantly stronger sequence constraints at terminal regions compared to non-secreted proteins, providing empirical support for the biological relevance of terminal signals. Experimental results demonstrate that TermSE achieves superior performance over existing methods in both cross-validation and independent test settings. Furthermore, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) [39] interpret model predictions, confirming that TermSE learns to focus on biologically meaningful terminal patterns specific to each secreted effector type, underscoring its utility for secreted effector discovery.

II. RESULTS

A. Overview of TermSE

We developed TermSE, a terminal signal-aware deep learning framework for multi-class secreted effector identification. The model architecture is designed based on the biological prior that

secretion systems recognize their substrates through specific signal sequences located at protein termini. As illustrated in Fig. 1A, TermSE consists of three parallel feature extraction branches followed by a fusion module and a cosine-normalized classifier.

Leveraging the representation capacity of pLMs, TermSE processes input sequences through ProtT5 to generate per-residue embeddings. These pLM-derived embeddings capture contextual sequence patterns through self-supervised learning on large-scale protein databases, providing informative representations that serve as the foundation for all downstream feature extraction. The embeddings are then processed through three parallel branches: the first branch applies mean pooling to obtain a global sequence representation, which is subsequently transformed through a global encoder consisting of a linear layer with batch normalization, ReLU activation, and dropout (Fig. 1B); the second and third branches extract the N-terminal and C-terminal segments, respectively, and process them through 1D CNNs to capture terminal signal patterns. As shown in Fig. 1B, each terminal CNN branch employs a 1D CNN layer followed by batch normalization, ReLU activation, and global max pooling to extract discriminative features from terminal regions.

Features from all three branches are concatenated and passed through a fusion layer. To address the severe class imbalance among different secreted effector types, TermSE employs a

cosine-normalized classifier (Fig. 1C) instead of a standard

linear classifier. This classifier computes the cosine similarity

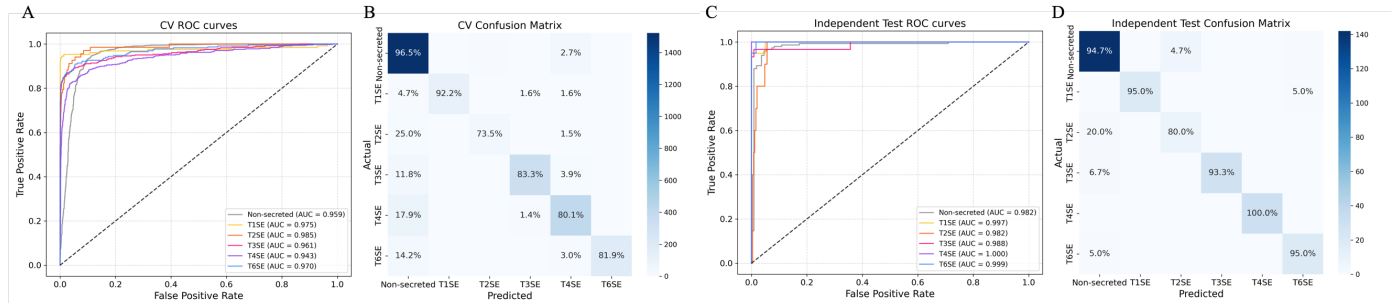


Fig. 3. Performance evaluation of TermSE on cross-validation and independent test sets. (A, C) Receiver operating characteristic (ROC) curves showing the classification performance for six classes on the 5-fold cross-validation (CV) and the independent test set, respectively. (B, D) Confusion matrices for the CV and independent test sets. The values on the main diagonal represent the recall for each class.

between L2-normalized feature vectors and learnable class weight vectors, scaled by a temperature parameter, effectively eliminating feature magnitude bias that would otherwise favor majority classes. The final output is a six-class probability distribution corresponding to five secreted effector types and non-secreted effectors.

B. PSSM-based analysis reveals secretion signal enrichment at protein termini

To test whether secretion signals are preferentially localized to protein termini, we quantified position-specific sequence constraints using PSSM-derived metrics, comparing terminal windows against mid-segment and global sequence backgrounds (Fig. 2). For N-terminal regions, both entropy-based (ΔH ; Fig. 2A, C) and preference-strength contrasts (ΔM ; Fig. 2E, G) showed consistently positive values for T1SE–T4SE and T6SE, with most classes exhibiting highly significant within-class deviations from zero ($p < 0.0001$). This indicates that N-terminal regions are more information-rich than either mid-segment or full-length backgrounds. Non-secreted proteins, by contrast, showed substantially weaker effects, with ΔH and ΔM values centered near zero and smaller effect sizes, demonstrating that pronounced N-terminal enrichment is characteristic of secreted effectors rather than a general property of all sequences.

C-terminal enrichment was present but more variable (Fig. 2B, D, F, H). Several secreted effector types showed positive and significant C-terminal entropy and preference indices, indicating signal enrichment relative to internal regions. However, effect sizes were generally smaller than those at the N-terminus, and some classes showed non-significant deviations from zero, particularly in mid-segment comparisons. This suggests that while N-terminal enrichment predominates, C-terminal regions contribute additional functional information in specific secreted effector types.

Between-class comparisons confirmed type-specific terminal signatures. Welch ANOVA yielded extremely small p-values below 10^{-10} for all N-terminal contrasts, reflecting robust differences among secreted effector types in both enrichment magnitude and constraint strength. C-terminal indices showed weaker but still significant between-class separation against the global background, while mid-segment comparisons did not reach significance. These results establish that secreted

effectors concentrate distinct, type-specific signals at their termini, with N-terminal regions exhibiting particularly strong and discriminative enrichment relative to internal and global sequence backgrounds. This provides direct empirical support for the biological relevance of terminal signals in secreted effector recognition.

C. TermSE achieves robust performance in multi-class secreted effector identification

To evaluate the classification performance of TermSE, we first conducted 5-fold cross-validation (CV) on the training set comprising 2,918 protein sequences. As shown in Fig. 3A, TermSE achieved excellent discriminative ability across all six classes, with the area under the receiver operating characteristic curve (AUC) ranging from 0.943 to 0.985. Notably, T2SE achieved the highest AUC of 0.985, while T4SE showed the lowest but still competitive AUC of 0.943. The confusion matrix further demonstrated high classification accuracy (Fig. 3B). Non-secreted proteins were classified with 96.5% recall, demonstrating the model's robust ability to distinguish secreted effectors from non-secreted proteins. Among the secretion systems, T1SE and T3SE achieved recall values of 92.2% and 83.3%, respectively, while T2SE showed relatively lower recall of 73.5%, likely attributable to its limited sample size of only 68 sequences in the dataset.

To further assess the generalization capability of TermSE, we evaluated the model on an independent test set. TermSE exhibited even stronger performance, with AUC values for all secretion systems exceeding 0.982, and notably reaching 1.000 for T4SE and 0.999 for T6SE (Fig. 3C). The confusion matrix (Fig. 3D) showed that T4SE was classified with a perfect 100.0% recall, while T1SE, T2SE, T3SE, and T6SE achieved 95.0%, 80.0%, 93.3%, and 95.0% recall, respectively. The non-secreted class maintained high recall at 94.7%, with minor misclassifications primarily attributed to T2SE.

To better understand how TermSE learns discriminative features, we performed UMAP analysis on the independent test set, comparing the input embeddings with the learned representations from TermSE (Supplementary Fig. S1). The initial feature space showed significant overlap among different secreted effectors and non-secreted proteins, whereas the learned representations from TermSE exhibited clear clustering patterns, with each secreted effector type forming distinct and

well-separated groups. This visualization demonstrates that

TermSE successfully transforms the initial sequence

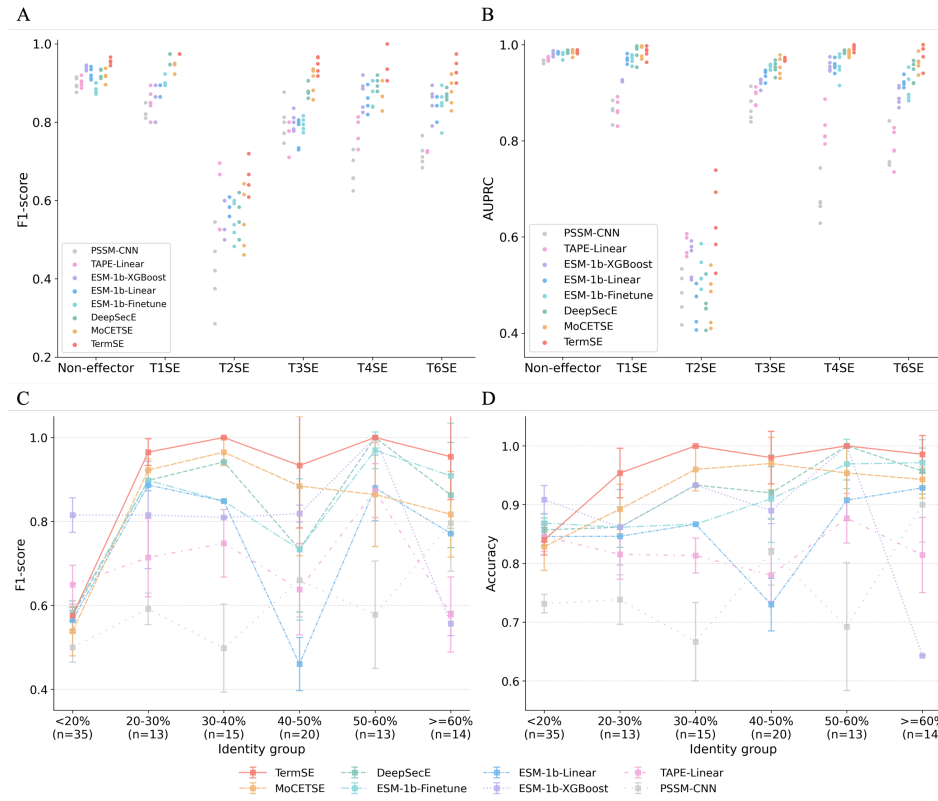


Fig. 4. Performance comparison of TermSE with existing methods. (A) F1-score and (B) AUPRC for each class on the independent test set. Each point represents a model trained from one fold of 5-fold cross-validation. (C) F1-score and (D) accuracy across different sequence identity groups on the independent test set. The number of proteins in each group is indicated in parentheses.

embeddings into a more discriminative feature space by learning terminal signal patterns specific to each secreted effector type.

Overall, these results demonstrate that TermSE generalizes well to unseen proteins and learns highly discriminative representations for secreted effector identification. The model achieves robust multi-class performance, with remaining misclassifications concentrated in biologically plausible borderline cases where terminal signals may be weak or partially overlapping.

D. TermSE outperforms state-of-the-art methods in benchmark comparisons

TABLE I

PERFORMANCE COMPARISON OF TERMSE WITH EXISTING METHODS.

Method	ACC		F1-score		AUPRC	
	CV	Ind	CV	Ind	CV	Ind
PSSM-CNN	0.799	0.822	0.712	0.724	0.752	0.774
TAPE-Linear	0.816	0.838	0.764	0.770	0.802	0.822
ESM-1b-XGBoost	0.869	0.887	0.809	0.816	0.865	0.872
ESM-1b-Linear	0.876	0.870	0.841	0.810	0.880	0.871
ESM-1b-Finetune	0.878	0.850	0.846	0.808	0.887	0.883
DeepSecE	0.883	0.898	0.848	0.849	0.892	0.885
MoCETSE	0.888	0.889	0.852	0.843	0.893	0.891
TermSE	0.899	0.939	0.880	0.904	0.907	0.924

Note: CV denotes 5-fold cross-validation; Ind denotes independent test set. Bold values indicate the best performance for each metric.

To validate the superiority of TermSE, we conducted a

comprehensive benchmark comparison against several existing methods, including PSSM-CNN, TAPE-Linear, ESM-1b-XGBoost, ESM-1b-Linear, ESM-1b-Finetune, DeepSecE, and MoCETSE. As summarized in Table I, TermSE achieved the best overall performance across all evaluation metrics. In cross-validation, TermSE attained the highest accuracy of 0.899, F1-score of 0.880, and AUPRC of 0.907, outperforming the second-best method MoCETSE by 1.1%, 2.8%, and 1.4%, respectively. On the independent test set, TermSE demonstrated even stronger performance, achieving the highest accuracy of 0.939, F1-score of 0.904, and AUPRC of 0.924, outperforming MoCETSE by 5.0%, 6.1%, and 3.3%, respectively.

The comparison between cross-validation and independent test performance highlights the superior generalization capability of TermSE. While MoCETSE exhibited competitive performance during cross-validation, its F1-score decreased from 0.852 to 0.843 on the independent test set, whereas TermSE improved from 0.880 to 0.904. This stability suggests that the terminal signal-aware architecture of TermSE effectively captures universal biological signatures rather than overfitting to dataset-specific patterns. Notably, several pLM-based methods including ESM-1b-Linear and ESM-1b-Finetune showed performance degradation on the independent test set, indicating potential overfitting, while TermSE consistently improved. The detailed per-method performance distributions are provided in Supplementary Fig. S2, showing that TermSE achieves not only higher mean performance but

also lower variance compared to most competing methods, indicating more stable and reliable predictions across different

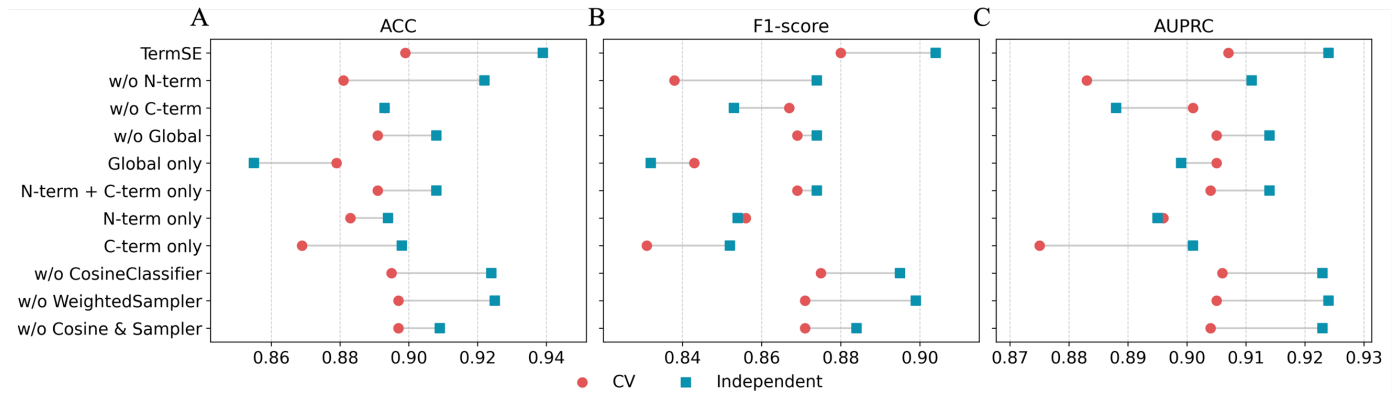


Fig. 5. Ablation study of TermSE. Performance comparison of different model variants in terms of (A) accuracy, (B) F1-score, and (C) AUPRC. Each dumbbell plot shows the CV (red circle) and independent test (blue square) performance, with connecting lines indicating the performance gap between the two evaluation settings.

evaluation settings.

To further evaluate model effectiveness across diverse secreted effectors, we analyzed the class-specific performance on the independent test set (Fig. 4A-B). For each method, we evaluated the five models trained from 5-fold cross-validation on the independent test set, allowing us to assess both performance and stability. TermSE achieved the best or near-best performance across all six categories in terms of both F1-score and AUPRC. For the challenging T2SE class, which contains only 68 samples, most methods exhibited substantially lower and highly variable performance, while TermSE achieved consistently high performance on this minority class. For T3SE and T4SE, which represent the largest secreted effector classes, TermSE achieved robust classification performance, consistently ranking among the top methods. As shown in Fig. 4A-B, TermSE maintained stable performance across all secreted effector types with low variance among the five models, whereas other methods showed greater variability, particularly for the underrepresented T2SE class. Notably, although some methods achieved higher scores on specific classes, they exhibited inconsistent performance across different secretion types, whereas TermSE provided more balanced and reliable predictions overall.

Overall, these results demonstrate that TermSE consistently outperforms existing methods in multi-class secreted effector classification, with particularly strong generalization capability and balanced performance across all secreted effector types.

E. TermSE demonstrates high robustness across varying sequence identities

To rigorously evaluate whether the predictive power of TermSE stems from learning universal biological patterns rather than relying on high sequence similarity to the training set, we performed a stratified performance analysis based on sequence identity. The independent test set comprises secreted effectors with a wide range of sequence identities to the training data. Specifically, 110 secreted effectors were divided into six groups according to their maximum sequence identity (<20% (n=35), 20-30% (n=13), 30-40% (n=15), 40-50% (n=20), 50-

60% (n=13), and $\geq 60\%$ (n=14)).

As shown in Fig. 4C-D, TermSE demonstrated superior performance across most sequence identity groups in terms of both F1-score and accuracy. For secreted effectors with extremely low sequence identity below 20%, which represent the most challenging cases for generalization, all methods showed relatively lower performance, with TermSE achieving comparable results to other top-performing methods in this difficult regime. As the sequence identity increased beyond 20%, TermSE emerged as the consistently top-performing model across all remaining groups. For proteins with sequence identity of 30-40% and 50-60%, TermSE achieved near-perfect performance, significantly surpassing all baseline methods. In the 20-30% and $\geq 60\%$ identity groups, TermSE also maintained excellent performance, consistently ranking at the top. Notably, in the 40-50% identity group, most methods experienced dramatic performance declines, likely due to the heterogeneous nature of proteins in this transitional range. While TermSE also showed reduced performance in this challenging group, it still substantially outperformed most competing methods and exhibited lower variance. Importantly, TermSE demonstrated lower variance across the five cross-validation models compared to most competing methods, particularly in groups with sequence identity above 20%.

These results demonstrate that TermSE maintains strong and stable predictive performance across a wide range of sequence identity levels, indicating that its terminal signal-aware architecture captures conserved biological signatures rather than relying on sequence homology to the training data.

F. Ablation study

To investigate the contribution of each component in TermSE, we conducted a comprehensive ablation study by systematically removing or replacing individual modules (Fig. 5). The following model variants were designed:

- w/o N-term: the model without N-terminal features.
- w/o C-term: the model without C-terminal features.
- w/o Global: the model without global embedding.

- Global only: the model using only global embedding,
- N-term + C-term only: the model using only terminal features, without global embedding.
- N-term only: the model using only N-terminal features.
- C-term only: the model using only C-terminal features.
- w/o CosineClassifier: the model replacing cosine-normalized classifier with standard linear classifier.
- w/o WeightedSampler: the model without weighted sampling.
- w/o Cosine & Sampler: the model without both cosine classifier and weighted sampling.

The terminal signal features proved to be the most critical component for model performance. Using only global embedding without terminal features (Global only) resulted in substantial performance degradation across all metrics, particularly on the independent test set. Conversely, using only terminal features without global embedding (N-term + C-term only) maintained competitive performance, though slightly lower than the full model, demonstrating that terminal signals capture the primary discriminative information for secreted effector classification.

The N-terminal and C-terminal features showed asymmetric contributions. Removing N-terminal features (w/o N-term) caused a more pronounced performance drop compared to removing C-terminal features (w/o C-term), particularly evident in the independent test F1-score. This observation aligns with the biological understanding that N-terminal signal peptides play a crucial role in protein secretion for certain secretion systems such as T2SS and T3SS. When examining single-terminal models, N-term only outperformed C-term only across all metrics, further confirming the greater importance of N-terminal signals in this classification task.

The class imbalance handling strategies also contributed substantially to model performance. Removing the cosine-normalized classifier (w/o CosineClassifier) led to noticeable performance drops, particularly in F1-score and AUPRC, indicating its effectiveness in mitigating feature magnitude bias across imbalanced classes. Similarly, removing the weighted sampling (w/o WeightedSampler) resulted in decreased performance, especially on the independent test set. When both strategies were removed simultaneously (w/o Cosine & Sampler), the model exhibited the most pronounced degradation among the class imbalance ablations, demonstrating the synergistic effect of these two components in handling severe class imbalance.

The ablation study confirms that each component of TermSE contributes to the overall performance, with terminal signal features serving as the primary driver and class imbalance strategies ensuring balanced learning across all secreted effector types. We evaluated seven different window sizes of 30, 50, 75, 100, 125, 150, and 200 residues. The results revealed that a window size of 100 residues achieved optimal performance, attaining the highest independent test accuracy of 0.939, F1-score of 0.904, and AUPRC of 0.924. Smaller window sizes of 30, 50, and 75 residues showed slightly lower

without any terminal features.

performance, likely due to incomplete capture of secretion signal regions for certain effector types. Conversely, larger window sizes of 125, 150, and 200 residues did not yield further improvements and introduced noise from non-signal sequences. These results support our choice of 100 residues as the default window size, which effectively balances comprehensive signal capture with noise avoidance.

G. Interpretability analysis confirms capture of biologically consistent terminal patterns

To validate that TermSE captures biologically meaningful information rather than relying on spurious correlations, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) [40] to the CNN subnetworks. Grad-CAM scores were computed with respect to the corresponding class logit and averaged across correctly classified proteins in the independent test set, yielding class-specific importance profiles along the N- and C-terminal sequences.

As shown in Fig. 6A-B, T2SE and T3SE exhibit distinct N-terminal signal localization. The N-terminal CNN branch shows a sharp importance peak within the first 20-30 residues, followed by a rapid decay toward the sequence interior, while C-terminal profiles remain relatively diffuse with moderate fluctuations. This pattern indicates that the model focuses on short N-terminal segments, aligning with known N-terminal secretion signals in these systems. In contrast, T4SE displays a pronounced C-terminal preference (Fig. 6C), where N-terminal importance is distributed without a dominant peak, while C-terminal scores increase progressively toward the sequence end, reaching maximum values within the last 30 residues, consistent with characterized C-terminal export signals. Systems with more complex signal architectures, such as T1SE and T6SE, show broader importance distributions (Supplementary Fig. S3), with N-terminal importance spanning residues 25-100 rather than concentrating at the extreme terminus, and C-terminal profiles exhibiting multiple distributed peaks. These results demonstrate that the model dynamically shifts attention according to the distinct secretion mechanisms of different systems, with decisions driven by biologically relevant terminal patterns.

To visualize learned terminal signals at residue-level resolution, we generated Grad-CAM-weighted sequence logos for a representative T4SE protein, the BepH effector from *Bartonella tribocorum*, randomly selected from the independent test set (Fig. 6D-E). Consistent with class-level profiles, the N-terminal logo shows high importance only at the first few residues, followed by an extended region with near-zero contribution. In contrast, the C-terminal logo exhibits a pronounced cluster of highly weighted residues within the last approximately 20 amino acids, with individual positions reaching near-maximum importance scores. This analysis confirms that for T4SE proteins, the model prioritizes compact C-terminal segments rather than distributing attention uniformly across the input sequence.

These results demonstrate that TermSE learns to focus on

secretion-related terminal signals rather than making position- unspecific predictions, providing interpretable evidence that the

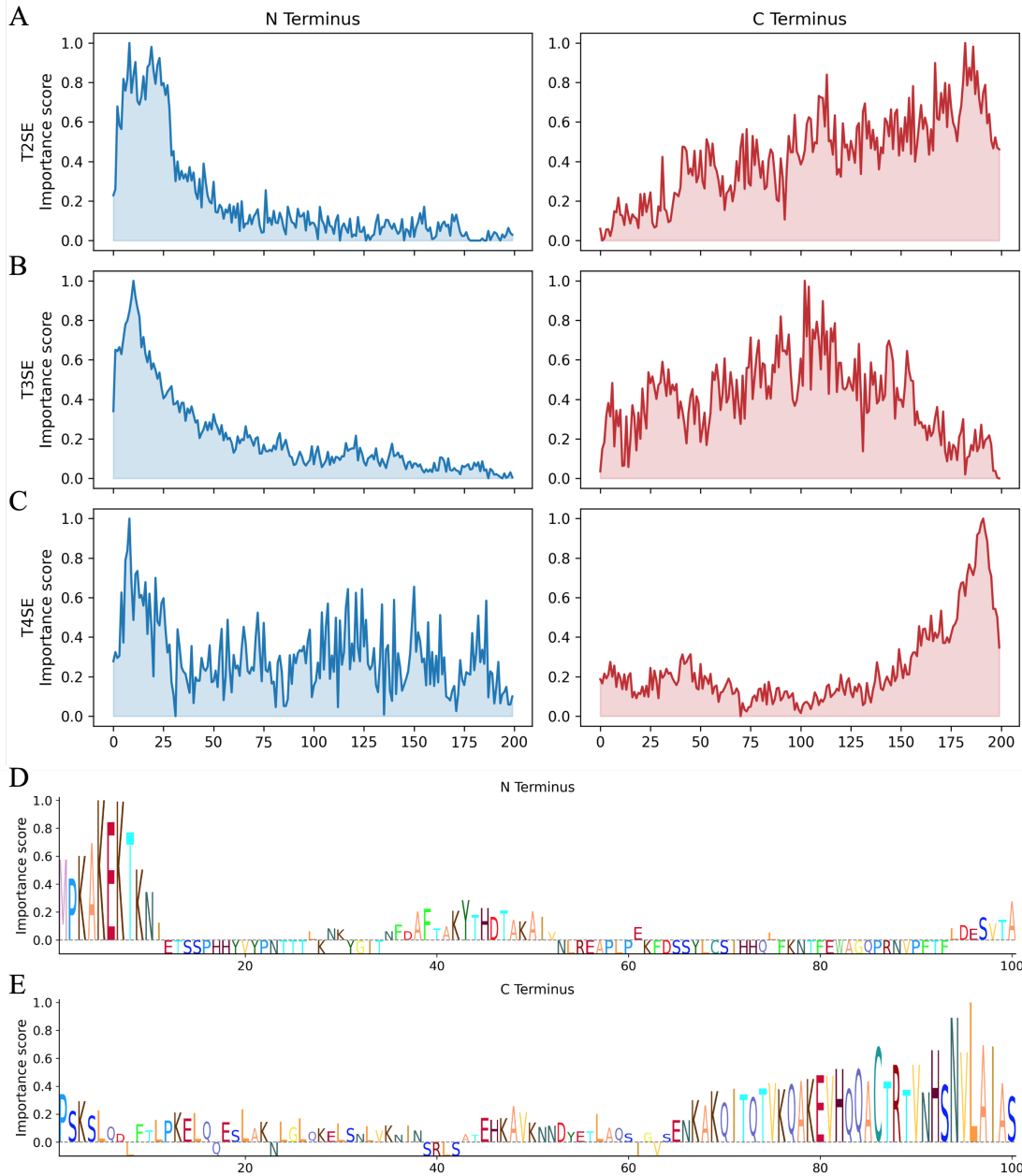


Fig. 6. Grad-CAM analysis of terminal region importance. (A-C) Importance profiles along 200-residue N-terminal (left) and C-terminal (right) windows for T2SE, T3SE, and T4SE proteins, averaged across correctly classified sequences in the independent test set. Importance scores were normalized to the range of 0-1 for each terminal region. (D-E). Grad-CAM-weighted sequence logos of a representative T4SE protein BepH, showing residue-level importance for the N-terminal region (D, first 100 residues) and C-terminal region (E, last 100 residues).

model captures biologically meaningful patterns.

III. MATERIALS AND METHODS

A. Dataset

We utilized the curated multi-class secreted protein dataset from DeepSecE [33], which integrates experimentally validated effectors across multiple secretion system types. The training dataset comprises 1,341 secreted proteins, including 128 T1SEs, 68 T2SEs, 406 T3SEs, 507 T4SEs, and 232 T6SEs, along with 1,577 nonsecreted proteins from Gram-negative bacteria. The independent test set contains 110 secreted proteins (20 T1SEs,

10 T2SEs, 30 T3SEs, 30 T4SEs, and 20 T6SEs) and 150 nonsecreted proteins. This dataset was originally compiled from multiple public databases, including SecReT4 [15] SecReT6 [16] and BastionHub [14], with negative samples derived from annotated nonsecreted proteins in UniProt [40] and homologous sequences from pathogenic strains. To ensure data quality and avoid information leakage, CD-HIT [41] was applied to remove redundant sequences with >60% sequence identity within the training set and between training and test sets.

B. Multi-view Sequence Representation

To capture complementary sequence cues relevant to secretion effector prediction, we constructed a multi-view sequence representation composed of three sequence-derived components, including a global semantic feature, an N-terminal local feature, and a C-terminal local feature. Starting from residue-level embeddings generated by ProtT5, these three views were separately encoded and subsequently fused into a unified representation for downstream classification.

Given a protein sequence of length L , ProtT5 produces a residue-level embedding matrix:

$$E \in \mathbb{R}^{L \times 1024}, \quad (1)$$

where $e_i \in \mathbb{R}^{1024}$ denotes the embedding of residue i . Based on this matrix, we derived three complementary views to characterize the protein sequence at both global and terminal levels. The global view was obtained by averaging all residue embeddings:

$$v_{\text{global}} = \frac{1}{L} \sum_{i=1}^L e_i. \quad (2)$$

Since bacterial secretion signals are typically enriched at the sequence termini, we explicitly defined terminal windows of length W to capture position-specific signal features. The N-terminal and C-terminal embedding windows are defined as:

$$E_N = [e_1, \dots, e_{\min(L, W)}], \quad (3)$$

$$E_C = [e_{\max(1, L-W+1)}, \dots, e_L]. \quad (4)$$

For sequences shorter than W ($L < W$), zero padding is applied; for longer sequences, only the first W residues (N-terminal) or last W residues (C-terminal) are retained.

C. The Model Architecture

We proposed TermSE, a terminal signal-aware deep learning framework that explicitly captures secretion-related signals at protein termini. As illustrated in Fig. 1, the model integrates N-terminal and C-terminal sequence features with global sequence representations through parallel processing branches, followed by a unified feature projection and a cosine-based classification module.

1) Signal-Aware Terminal Encoder

To detect localized signal motifs at the sequence ends, we designed identical but independent 1D CNNs for both N- and C-terminal windows. The extracted N-terminal and C-terminal feature matrices ($E_N, E_C \in \mathbb{R}^{W \times 1024}$) are processed to capture position-specific secretion signals. Specifically, a 1D CNN layer with a fixed kernel size ($k=21$) is employed, followed by Batch Normalization, ReLU activation, and Global Max Pooling to extract the most salient signal features. Formally, the operation is defined as:

$$H = \text{ReLU}(\text{BatchNorm}(\text{Conv1d}(E))), \quad (5)$$

$$v = \text{GlobalMaxPool}(H), \quad (6)$$

where v_N and v_C denote the final condensed representations for the N-terminal and C-terminal regions, respectively.

2) Global Sequence Encoding

Parallel to the terminal signal extraction, the global sequence vector v_{global} is processed through a Multi-Layer Perceptron

(MLP) block. This block consists of a linear transformation, Batch Normalization, ReLU activation, and Dropout regularization, projecting the 1024-dimensional raw global feature into a compact latent space:

$$v_G = \text{MLP}_{\text{global}}(v_{\text{global}}), \quad (7)$$

where v_G represents the encoded global semantic features, capturing the overall functional context of the protein.

3) Feature Fusion and Cosine Classification

The features from all three branches, including N-terminal signals, C-terminal signals, and global sequence context, are concatenated into a unified vector: $v_{\text{concat}} = [v_N, v_C, v_G]$. This fused representation is then fed into a feature projection module, which comprise a fully connected layer, Batch Normalization, ReLU, and Dropout, to yield the final fused representation v_{fusion} .

Instead of a standard linear classifier, TermSE utilizes a Cosine Classifier with temperature scaling to compute the similarity between the fused feature and learnable class weight vectors. This approach enhances the intra-class compactness and inter-class separability. The predicted probability distribution over the six secretion system classes is formulated as:

$$\hat{y} = \text{Softmax}(\tau \cdot \cos(v_{\text{fusion}}, W_c)), \quad (8)$$

where W_c denotes the weight matrix of the classes, $\cos(\cdot, \cdot)$ computes the cosine similarity, τ is a learnable temperature scaling parameter, and $\hat{y}$ is the final predicted probability. The detailed architecture parameters are summarized in Supplementary Table S2.

D. Model Training and Optimization

Five-fold stratified cross-validation was used to develop the TermSE model for secreted protein prediction, with class proportions approximately preserved across different folds. In each round, four folds were used for training and the remaining fold was used for validation.

To alleviate the inherent class imbalance in the dataset, we employed a dual rebalancing strategy at both the data and objective levels during training. At the data level, mini-batches were generated using a weighted random sampling strategy, such that samples from underrepresented classes were presented more frequently. At the objective level, class-dependent weights were incorporated into the loss function. For class c containing n_c samples, the corresponding weight was defined as:

$$w_c = \sqrt{\frac{N}{C \cdot n_c}}, \quad (9)$$

where N is the number of training samples in the current fold and C is the number of classes. This square-root inverse-frequency formulation provided a moderate correction for class imbalance, thereby increasing the contribution of minority classes without introducing overly large updates during optimization.

The model was trained using a class-weighted cross-entropy loss coupled with label smoothing. Label smoothing was

explicitly introduced to reduce overconfident predictions and improve generalization in this multi-class setting. The loss function was defined as:

$$\mathcal{L} = - \sum_{c=1}^C w_c \tilde{y}_c \log p_c, \quad (10)$$

where p_c denotes the predicted probability for class c , w_c is the class-specific weight, and $\tilde{y}_c$ represents the smoothed target distribution.

Model parameters were optimized using the AdamW optimizer with an initial learning rate 3×10^{-4} and a weight decay of 5×10^{-3} . A cosine annealing schedule was applied to gradually decrease the learning rate during the training process. Each model was trained for up to 80 epochs, and an early stopping strategy was adopted with a patience of 15 epochs. Specifically, the model parameters achieving the highest macroscopic F1-score on the validation fold were retained to prevent overfitting and ensure balanced performance across all secretion classes. For the final evaluation, the five optimal models obtained from the cross-validation were independently applied to the independent test set, and their average performance metrics were reported.

E. PSSM-based terminal signal analysis

To test whether secretion-related signals are preferentially localized to the N- or C-terminal regions, we quantified position-specific evolutionary preferences using Position-Specific Scoring Matrices (PSSMs). We compared N- and C-terminal windows against both a central control segment of the same length and the whole-sequence background. For each protein sequence, $S = (a_1, a_2, \dots, a_L)$, we generated a PSSM using PSI-BLAST [42] against the Swiss-Prot database with three iterations and an E-value threshold of 0.001. The resulting matrix $P \in \mathbb{R}^{L \times 20}$ contains one row per sequence position and one column per amino acid type, where the entry $P_{i,a}$ represents the log-odds score reflecting the evolutionary preference for amino acid a at position i .

To obtain probabilistic preference profiles, each PSSM row was converted into a temperature-scaled softmax distribution

$$p_i(a) = \frac{\exp(\beta P_{i,a})}{\sum_b \exp(\beta P_{i,b})}, \quad (11)$$

with $\beta = 1$ in this study. The Shannon entropy at position i was then computed as:

$$H_i = - \sum_a p_i(a) \log(p_i(a) + \varepsilon), \quad (12)$$

where ε is a small constant for numerical stability. For a given region $\mathcal{R}$ (e.g. N-terminal window, C-terminal window, mid-segment, or the full sequence), we summarized positional selectivity by the mean entropy:

$$H_{\mathcal{R}} = \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} H_i. \quad (13)$$

Lower $H_{\mathcal{R}}$ indicates higher information content and stronger positional constraints. Independently, we quantified the strength of evolutionary preference as the maximum PSSM score at each position $M_i = \max_a P_{i,a}$, and defined the regional mean:

$$M_{\mathcal{R}} = \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} M_i. \quad (14)$$

For each sequence, we defined N- and C-terminal windows

as the first and last W residues, respectively, a central control window (“Mid”) of length W centered in the sequence, and the full-length region (“Global”). We then computed two types of contrast indices:

- Terminus vs mid-segment:

$$\begin{aligned} \Delta H_{NM} &= H_{\text{Mid}} - H_N, & \Delta H_{CM} &= H_{\text{Mid}} - H_C, \\ \Delta M_{NM} &= M_N - M_{\text{Mid}}, & \Delta M_{CM} &= M_C - M_{\text{Mid}}. \end{aligned} \quad (15)$$

- Terminus vs global background:

$$\begin{aligned} \Delta H_{Ng} &= H_{\text{Global}} - H_N, & \Delta H_{Cg} &= H_{\text{Global}} - H_C, \\ \Delta M_{Ng} &= M_N - M_{\text{Global}}, & \Delta M_{Cg} &= M_C - M_{\text{Global}}. \end{aligned} \quad (16)$$

Positive ΔH values indicate that terminal regions exhibit lower entropy (i.e., are more information-rich) than the corresponding control region, whereas positive ΔM values indicate stronger evolutionary preferences at the termini. These indices were used in statistical tests (Wilcoxon and Welch ANOVA) to quantify terminal signal enrichment across secretion systems.

F. Performance Assessment

To comprehensively evaluate the predictive performance of our model, we employed multiple classification metrics across both 5-fold cross-validation and independent test sets. The overall discriminative ability was assessed using macro-averaged prediction accuracy, F1-score, area under the receiver operating characteristic curve (AUC), and area under the precision-recall curve (AUPRC). These macro-averaged metrics provide an unbiased evaluation of model performance across all classes, which is particularly important for handling class imbalance. Accuracy and F1-score were computed using the following formulations:

$$ACC = \frac{TP+TN}{TP+FP+TN+FN}, \quad (17)$$

$$F1 - score = \frac{2TP}{2TP+FN+FP}, \quad (18)$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively. The F1-score provides a harmonic mean balancing precision and sensitivity. Additionally, ROC curves and confusion matrices were generated to visualize classification performance and facilitate intuitive interpretation of model predictions. All performance metrics were computed using the scikit-learn library in Python.

IV. DISCUSSION

In this work, we proposed TermSE, a terminal signal-aware deep learning framework for multi-class identification of Gram-negative bacterial secreted effectors across five major secreted effector types and non-secreted proteins. Motivated by the biological observation that bacterial secretion systems rely on distinct terminal signal patterns for substrate recognition, we designed a model that explicitly captures N-terminal and C-terminal sequence features using 1D CNNs applied to pLM embeddings, while integrating global sequence representations from ProtT5 through a fusion module. To address the severe class imbalance among different secretion system types, TermSE employs a cosine-normalized classifier combined with

balanced sampling strategy. To validate our terminal-focused design, we first performed PSSM-based analysis, which demonstrated that secreted effectors exhibit significantly stronger sequence constraints at terminal regions compared to non-secreted proteins, providing empirical support for the biological relevance of terminal signals. Comprehensive experiments demonstrated the superior performance of TermSE over seven state-of-the-art baseline methods in both cross-validation and independent test settings. Furthermore, we conducted stratified analysis across different sequence identity levels, showing that TermSE maintains strong generalization capability even for proteins with low sequence similarity to the training data. Ablation studies confirmed that each component contributes to the overall performance, with terminal signal features being the most critical. Finally, Grad-CAM interpretability analysis revealed that TermSE learns to focus on biologically meaningful terminal patterns specific to each secreted effector type, with distinct N-terminal or C-terminal preferences consistent with known secretion mechanisms. All these results indicate that TermSE can provide accurate secreted effector predictions as well as useful insights into understanding secretion signal mechanisms.

Compared with existing pLM-based methods, TermSE offers several advantages. First, TermSE explicitly models the biological prior that secretion signals are localized at protein termini, whereas previous methods such as DeepSecE and MoCETSE primarily rely on global sequence representations that may dilute these functionally critical signals. This terminal-focused design enables TermSE to capture more discriminative features for distinguishing between secreted effector types. Second, TermSE demonstrates superior generalization capability. While MoCETSE achieved competitive performance during cross-validation, its performance dropped notably on the independent test set, suggesting potential overfitting to training data. In contrast, TermSE maintained consistent performance across both evaluation settings, indicating that the terminal signal-aware architecture captures universal biological signatures rather than dataset-specific patterns. Third, TermSE effectively addresses the class imbalance challenge through the combination of cosine-normalized classifier and weighted sampling, enabling robust classification of underrepresented effector types such as T1SE and T2SE. Fourth, TermSE provides interpretable predictions through Grad-CAM analysis, allowing researchers to verify whether the model focuses on biologically relevant regions. This interpretability is particularly valuable for guiding downstream experimental validation and understanding the molecular basis of secretion system specificity.

Nevertheless, certain limitations still exist in the current version of TermSE. First, for proteins with extremely low sequence identity below 20% to the training data, all methods including TermSE showed relatively lower performance, representing a common challenge for generalization to highly divergent sequences. Second, the performance on T2SE classification remains relatively lower compared to other secreted effector types, likely due to the limited sample size of

only 68 sequences in the dataset. The scarcity of experimentally validated T2SE proteins poses challenges for learning robust discriminative features. Third, the current version of TermSE relies solely on sequence-derived features and does not incorporate protein structural information, which could potentially provide complementary signals for effector recognition. In the future, we will explore integrating structure-aware learning approaches and few-shot learning strategies to improve multi-class prediction and enable better generalization to novel effector types with scarce experimental data.

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